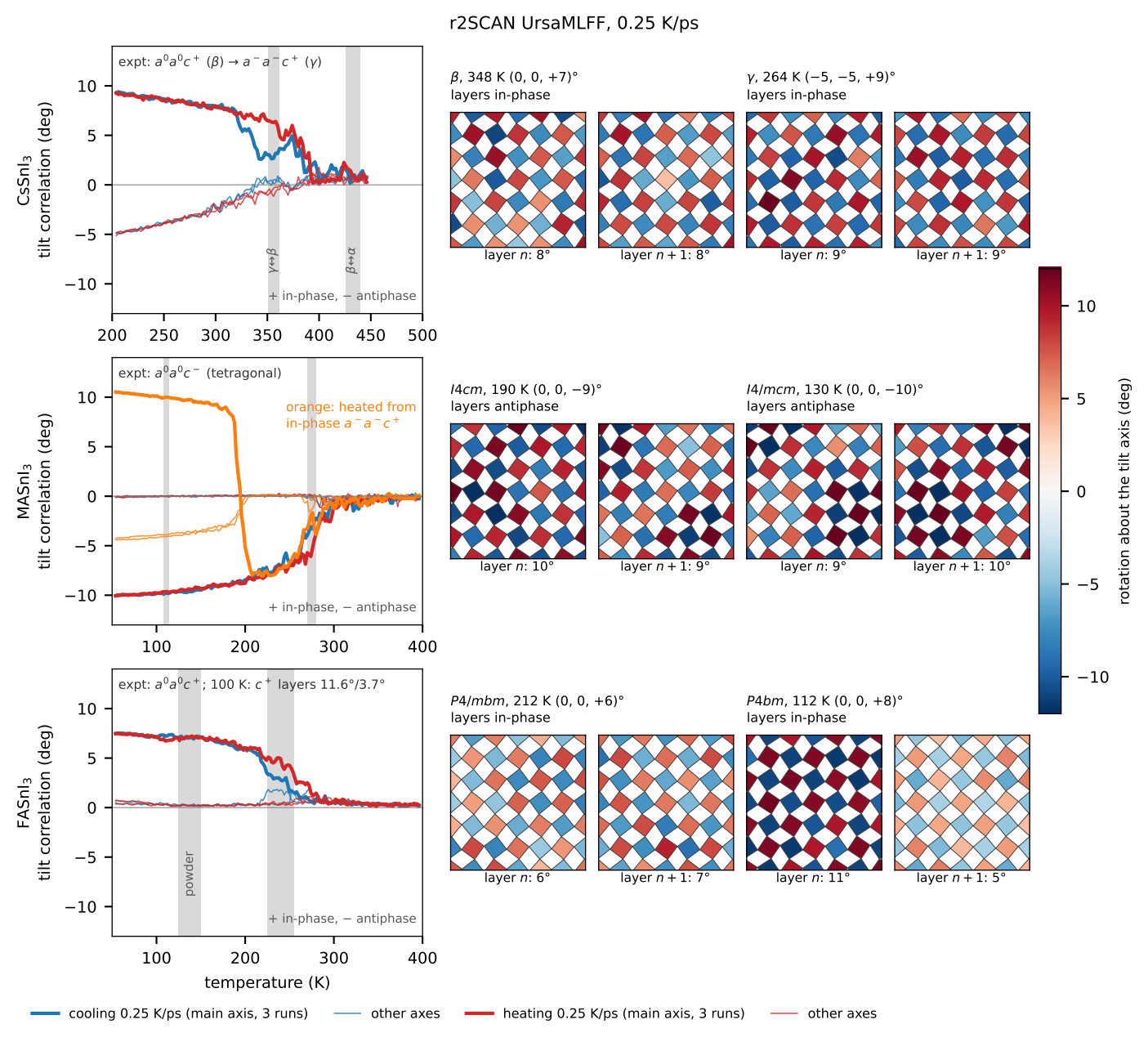

r2SCAN UrsaMLFF, 0.25 K/ps
expt: $a^0a^0c^+$ ($\beta$) → $a^-a^-c^+$ ($\gamma$)
$CsSnI_3$ tilt correlation (deg)
$\gamma \leftrightarrow \beta$
$\beta \leftrightarrow \alpha$
+ in-phase, − antiphase
$\beta$, 348 K (0, 0, +7)°
layers in-phase
$\gamma$, 264 K (−5, −5, +9)°
layers in-phase
layer $n$: 8°
layer $n+1$: 8°
layer $n$: 9°
layer $n+1$: 9°
expt: $a^0a^0c^-$ (tetragonal)
orange: heated from in-phase $a^-a^-c^+$
$MASnI_3$ tilt correlation (deg)
+ in-phase, − antiphase
I4cm, 190 K (0, 0, −9)°
layers antiphase
I4/mcm, 130 K (0, 0, −10)°
layers antiphase
layer $n$: 10°
layer $n+1$: 9°
layer $n$: 9°
layer $n+1$: 10°
expt: $a^0a^0c^+$; 100 K: $c^+$ layers 11.6°/3.7°
$FASnI_3$ tilt correlation (deg)
powder
+ in-phase, − antiphase
P4/mbm, 212 K (0, 0, +6)°
layers in-phase
P4bm, 112 K (0, 0, +8)°
layers in-phase
layer $n$: 6°
layer $n+1$: 7°
layer $n$: 11°
layer $n+1$: 5°
rotation about the tilt axis (deg)
temperature (K)
cooling 0.25 K/ps (main axis, 3 runs)
other axes
heating 0.25 K/ps (main axis, 3 runs)
other axes